Token-Generation Latency Benchmarking in LLaMA

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***Abstract*—Large Language Models (LLMs) are increasingly deployed in interactive applications where response speed directly affects usability. Unlike traditional batch inference pipelines, LLMs generate outputs autoregressively, making token-generation latency a central performance concern. This paper presents a benchmarking study of token-generation latency in a LLaMA-family model. The study focuses on Time to First Token (TTFT), Time Per Output Token (TPOT), throughput, and end-to-end latency under varying prompt and output lengths. The analysis shows that prompt length strongly affects first-token latency, while longer token sequences increase total completion time. These observations highlight the importance of evaluating LLM inference with multiple latency-oriented metrics rather than relying on throughput alone.**

I. Research Contributions

- Benchmark token-generation latency in LLaMA-family models.
- Analyze TTFT, TPOT, throughput, and end-to-end latency.
- Demonstrate the separation between prompt-processing and token-generation phases.
- Provide practical observations for real-world deployment.

II. Introduction

Large Language Models such as LLaMA are widely used in applications including chat systems, summarization tools, code assistants, and interactive reasoning platforms. Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothee Lacroix, Baptiste Roziere, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample introduced LLaMA as an efficient open-weight model family with strong performance characteristics [1].

At the same time, efficient serving of LLMs has become a major systems challenge. Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica proposed memory-efficient inference using PagedAttention, showing that serving efficiency depends heavily on KV-cache management [2]. Connor Holmes, Masahiro Tanaka, Michael Wyatt, Ammar Ahmad Awan, Jeff Rasley, Samyam Rajbhandari, Reza Yazdani Aminabadi, Heyang Qin, Arash Bakhtiari, Lev Kurilenko, and Yuxiong He further showed that prompt processing is a dominant source of latency in LLM inference [3].

Benchmarking studies have also emphasized the importance of prompt length and output length as key performance variables. Krishna Teja Chitty-Venkata, Siddhisanket Raskar, Bharat Kale, Farah Ferdaus, Aditya Tanikanti, Ken Raffenetti, Valerie Taylor, Murali Emani, and Venkatram Vishwanath highlighted that these variables are central to understanding inference behavior across systems [4]. Zhengqing Yuan, Weixiang Sun, Yixin Liu, Huichi Zhou, Rong Zhou, Yiyang Li, Zheyuan Zhang, Wei Song, Yue Huang, Haolong Jia, Keerthiram Murugesan, Yu Wang, Lifang He, Jianfeng Gao, Lichao Sun, and Yanfang Ye argued that LLM efficiency should be evaluated in a multi-dimensional manner rather than through a single performance number [5].

Additional work has reinforced the importance of fine-grained latency metrics. Hung-Yueh Chiang, Bokun Wang, and Diana Marculescu introduced ELANA to measure TTFT and TPOT explicitly [6]. Yang Cheng, Sisi Qin, Jing Qi, Feng Cao, and Dongwei Yan studied token-generation latency in service-based environments [7]. Patrick Wilhelm, Thorsten Wittkopp, and Odej Kao proposed energy-per-token as an additional efficiency-oriented metric [8]. Junyi Chen, Chuheng Du, Renyuan Liu, Shuochao Yao, Dingtian Yan, Jiang Liao, Shengzhong Liu, Fan Wu, and Guihai Chen demonstrated that scheduling policies affect responsiveness in practical LLM serving [9]. Minghao Shao, Abdul Basit, Ramesh Karri, and Muhammad Shafique provided a broad survey of LLM architectures, trends, benchmarks, and challenges [10].

Motivated by these works, this paper examines token-generation latency in LLaMA with a focus on how prompt length and output length influence user-facing performance.

III. Background

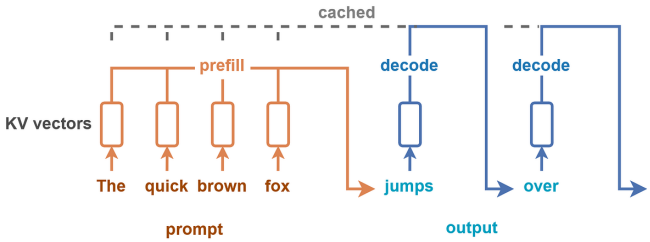

Fig. 1. Prefill and decode phases in LLaMA inference

LLaMA is a decoder-only transformer model that generates text autoregressively. Inference consists of two main stages. During the prefill phase, the model processes the full input

prompt and constructs the internal state required for generation. During the decode phase, output tokens are produced sequentially using cached representations from earlier computations.

This separation is important because different latency metrics are influenced by different stages. TTFT is strongly tied to prompt-processing cost, whereas later token generation depends more directly on decoding efficiency.

IV. Methodology

A. Experimental Setup

Experiments were conducted on a CPU using a smaller LLaMA-family model (TinyLlama) as a proxy due to hardware constraints. The model was executed using the HuggingFace Transformers framework with KV-cache enabled to optimize decoding efficiency.

All experiments were performed with a batch size of 1 to simulate real-time interactive workloads, where latency is more critical than throughput. Latency metrics were measured using high-resolution system timers, capturing the time from request submission to token generation.

Each configuration (prompt and output length combination) was executed five times, and the average values were reported to reduce measurement variance and improve reliability of results.

B. Variables

- Prompt Length: 32, 128, 256, 512 tokens
- Output Length: 32, 64, 128 tokens

C. Metrics

- Time to First Token (TTFT)
- Time Per Output Token (TPOT)
- End-to-End Latency
- Throughput (tokens/second)

V. Results

A. Prompt Length Impact

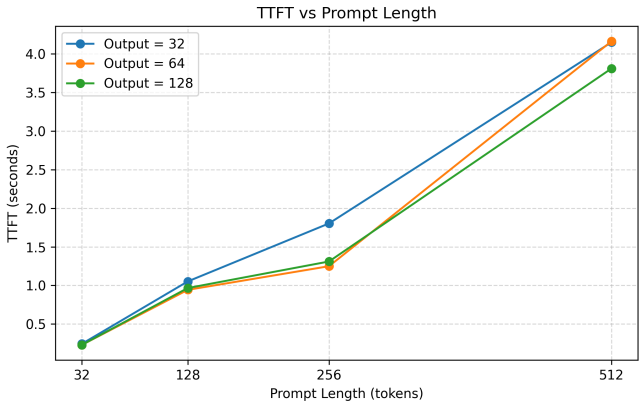

Fig. 2. TTFT increases with prompt length

The experimental results indicate that prompt processing contributes significantly to initial latency, while decoding contributes incrementally to total generation time. As prompt length increases from 32 to 512 tokens, TTFT increases noticeably due to the higher computational cost in the prefill phase.

This trend highlights that TTFT scales approximately with input size, as the model must process the entire prompt before generating the first token. These observations are consistent with prior findings in LLM inference studies [3], [6].

B. Output Length Impact

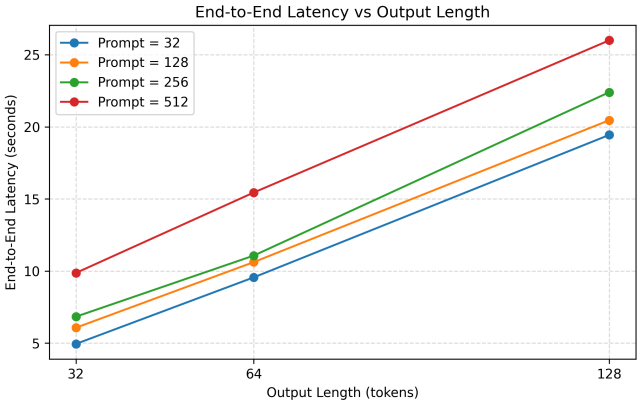

Fig. 3. End-to-end latency increases with token length

End-to-end latency increases linearly with output length due to the autoregressive nature of token generation. Each additional token introduces sequential computation, resulting in cumulative latency growth.

For example, increasing output length from 32 to 128 tokens results in a proportional increase in total completion time, demonstrating that decoding latency dominates long responses [4].

C. Throughput Analysis

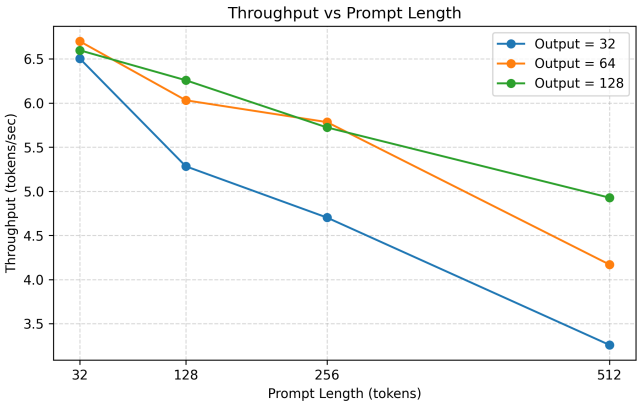

Fig. 4. Throughput variation across different token sizes

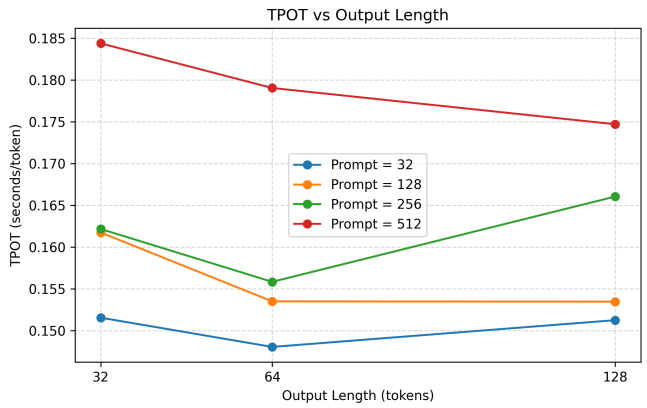

Fig. 5. TPOT remains relatively stable across output lengths

Throughput improves with larger workloads due to better hardware utilization and batching efficiency. However, higher throughput does not necessarily imply better user experience.

In several cases, configurations with higher throughput still exhibited increased TTFT, leading to delayed response initiation. This demonstrates that throughput alone is insufficient for evaluating interactive system performance and must be interpreted alongside latency metrics [8], [9].

VI. Discussion

The observed latency trends can be explained by the architectural properties of transformer-based LLMs. During the prefill phase, attention computation scales with the input sequence length, leading to increased memory access and computational overhead. This directly impacts TTFT, making it highly sensitive to prompt length.

In contrast, during the decode phase, the KV-cache enables reuse of previously computed attention states, reducing computation per token. As a result, TPOT remains relatively stable even as prompt length increases. This clearly demonstrates the separation between prefill and decode stages in LLM inference.

However, the autoregressive nature of token generation introduces a fundamental limitation: tokens must be generated sequentially, preventing parallelization. This leads to a linear increase in end-to-end latency as output length grows.

Another important observation is the trade-off between throughput and latency. While batching and larger workloads improve GPU utilization and increase throughput, they can delay the processing of individual requests, resulting in higher TTFT. This creates a tension between system-level efficiency and user-perceived responsiveness.

Therefore, optimizing LLM inference requires a balanced approach that considers both latency and throughput. Focusing on a single metric can lead to misleading conclusions, particularly in interactive applications where responsiveness is critical.

VII. Key Findings

Based on the analysis of token-generation latency in LLaMA models, the following findings are observed:

- **Prompt length has a significant impact on TTFT.** As the number of input tokens increases, the model requires more computation during the prefill phase, leading to higher Time to First Token. This confirms that prompt-processing cost dominates initial latency.
- **Output length directly affects end-to-end latency.** Longer output sequences increase total latency due to the sequential nature of autoregressive token generation. Each additional token introduces additional computation.
- **TTFT is more sensitive to input size than TPOT.** While TTFT increases noticeably with prompt length, the time per output token remains relatively more stable. This demonstrates the separation between prefill and decode phases.
- **Throughput alone does not represent user-perceived performance.** Even when throughput improves, users may still experience delays if TTFT is high. Therefore, latency and throughput must be considered together.
- **LLM inference performance is multi-dimensional.** A single metric is insufficient to evaluate performance. TTFT, TPOT, end-to-end latency, and throughput must all be analyzed to understand overall system behavior.

VIII. Conclusion

This study demonstrates that token-generation latency in LLaMA is influenced by both prompt length and output length. Prompt length significantly impacts first-token latency due to prefill computation, while output length determines total completion time due to sequential decoding.

The results further show that LLM inference performance cannot be accurately captured using a single metric. Instead, TTFT, TPOT, throughput, and end-to-end latency must be jointly analyzed to understand system behavior.

These insights are particularly relevant for designing and deploying real-time LLM applications, where responsiveness is critical. Future work can extend this study by incorporating additional factors such as hardware variations, batching strategies, and model size comparisons.

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